



BITS Pilani
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Economic Decision Making



Outline

- Design of A Solvent Recovery System (Ch. 3 of T2)
 - Problem Definition
 - Economic Potential
 - Process alternatives



Problem Definition

- As a part of a process design problem
 - Assume that there is a stream
 - Containing 10.3 mol/hr of acetone and 687 mol/hr of air
 - That is being fed to a flare system (to avoid air pollution)



Economic Potential (EP)

- Economic Potential (EP)

$$\text{EP} = \text{Product Value} - \text{Raw Material Cost}$$

- Since stream coming from the same process,
Raw material cost = 0
- Therefore

$$\text{EP} = \text{Product Value} - \text{Raw Material Cost}$$

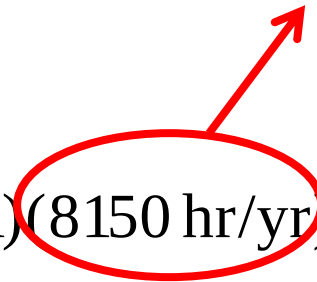
$$= \text{Product Value} - 0$$

$$= \text{Product Value}$$

$$= (10.3 \text{ mol/hr})(\text{Rs } 10.80/\text{lb})(58 \text{ lb/mol})(8150 \text{ hr/yr})$$

$$= \text{Rs. } 5.26 \text{ Cr/yr}$$

Operating hours





Question 1

How to recover acetone?



General Considerations: Process Alternatives

- Solvent recovery alternatives
 1. Condensation
 - a. High Pressure
 - b. Low temperature
 - c. Combination of both
 2. Absorption
 3. Adsorption
 4. A Membrane Separation System
 5. A Reaction Process (Acetone as raw material for a new product)



Question 2

Which is the cheapest alternative?



General Considerations: Process Alternatives

- If solute concentration (mole fraction) in a gas $< 5 \%$
 - Adsorption is the cheapest process
 - In the present case, it is $\approx 1.5 \%$
 $[10.3/(687+10.3) = 0.0147]$
- may opt for Adsorption
- However, many petroleum companies prefer to use
 - Condensation or absorption process



Contd..

- Judgment based on:

Concerning the use of technology where we have
great deal of experience

VS.

Using a technology where we have much less
experience (Relatively new technology)

Design of Gas Recovery System: Flowsheet



Douglas, J. M. Conceptual Design of Chemical Processes, 1988

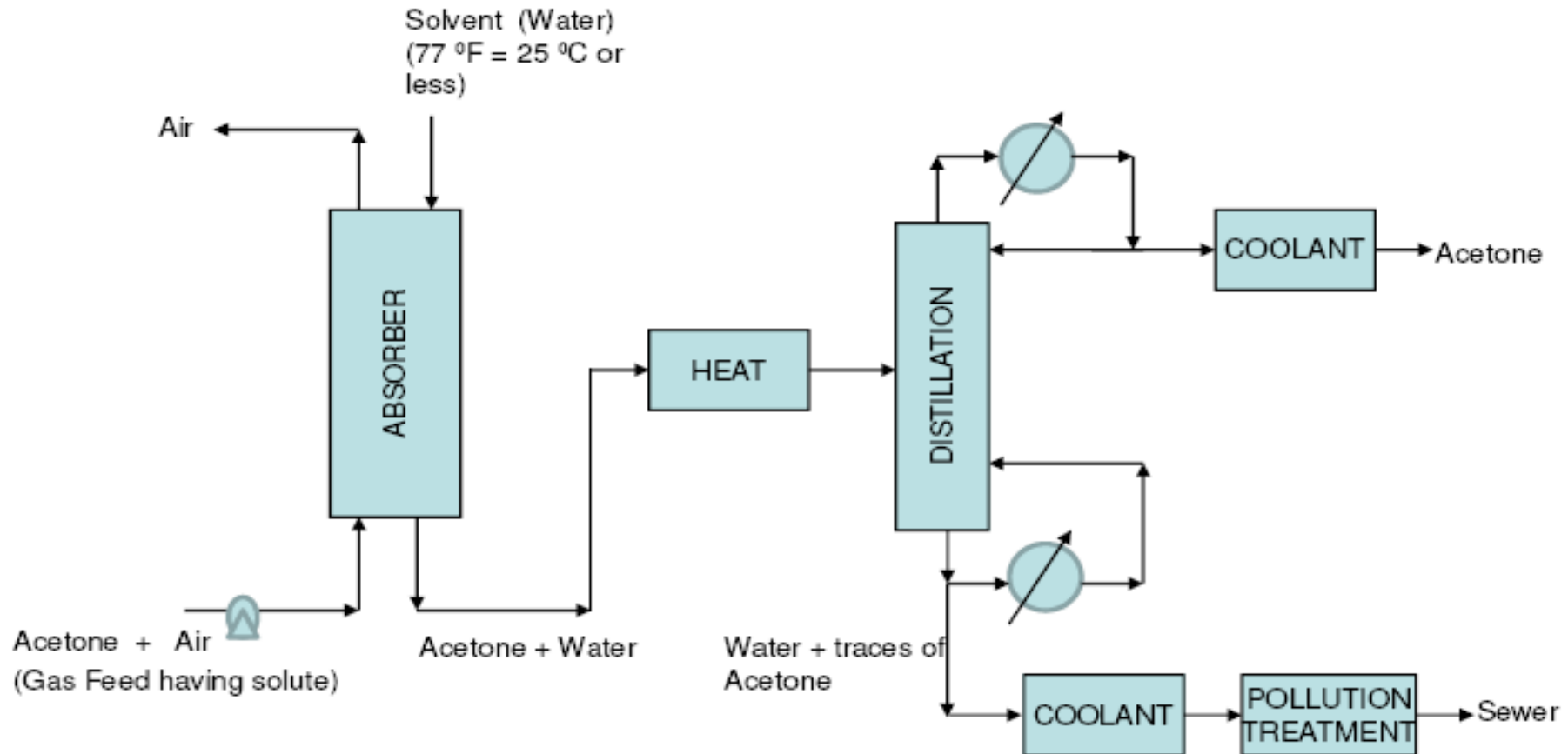


Fig. 1 Flow sheet of gas absorption

Alternate Flow sheet: Recycling of solvent

Douglas, J. M. Conceptual Design of Chemical Processes, 1988

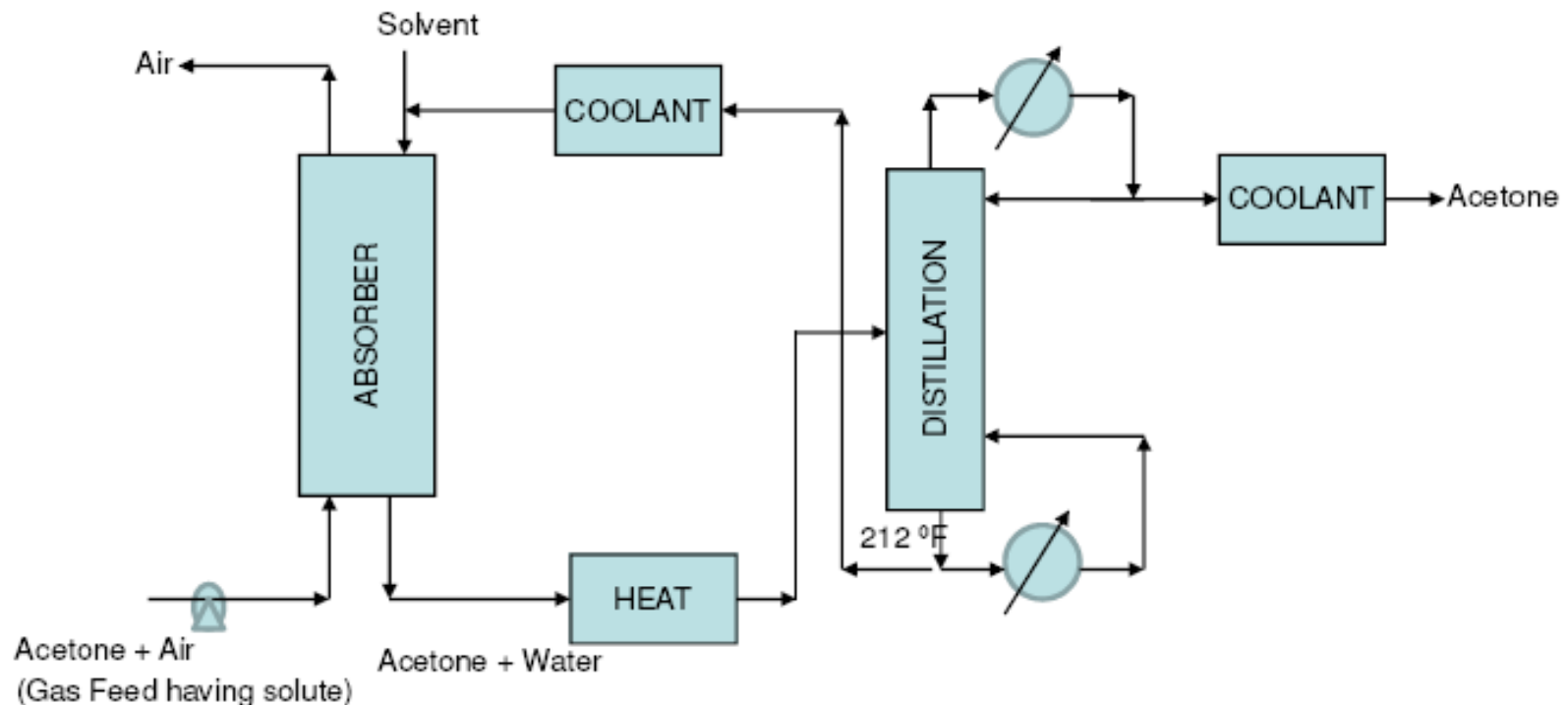


Fig. 2 Alternate flow sheet of gas absorption

Alternate Flow sheet: Recycling of solvent

Douglas, J. M. Conceptual Design of Chemical Processes, 1988

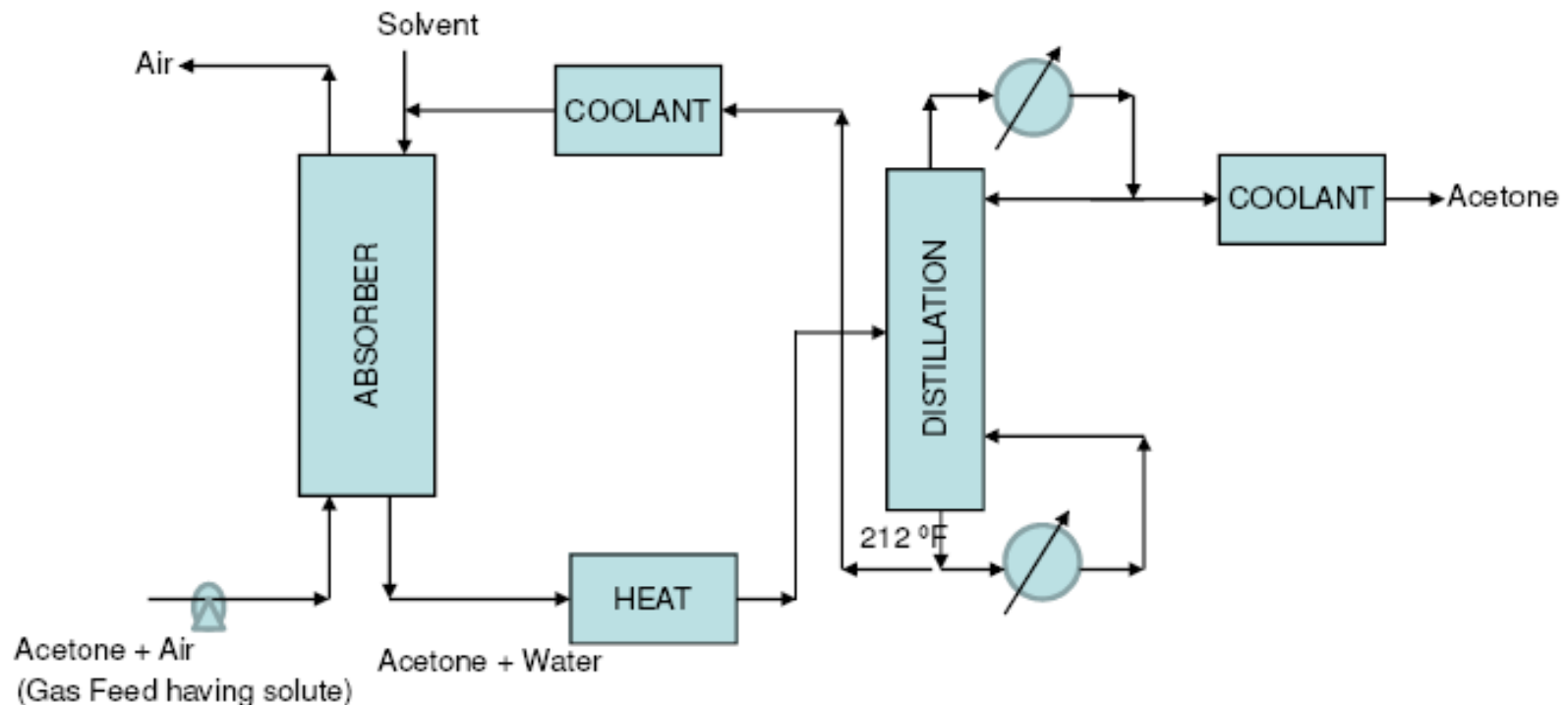


Fig. 2 Alternate flow sheet of gas absorption



Question 3

Whether discarding the process water, as shown in Fig. 1 can ever be justified even when a pollution treatment facility is available?



Contd..

- Check the temperature of the process water entering the gas absorber
- Cooling water is available from the cooling towers at **90 °F** (32 °C) (on the hot summer day)
- And that is must be returned to the cooling towers at a temperature less than **120 °F** (49 °C)



Contd..

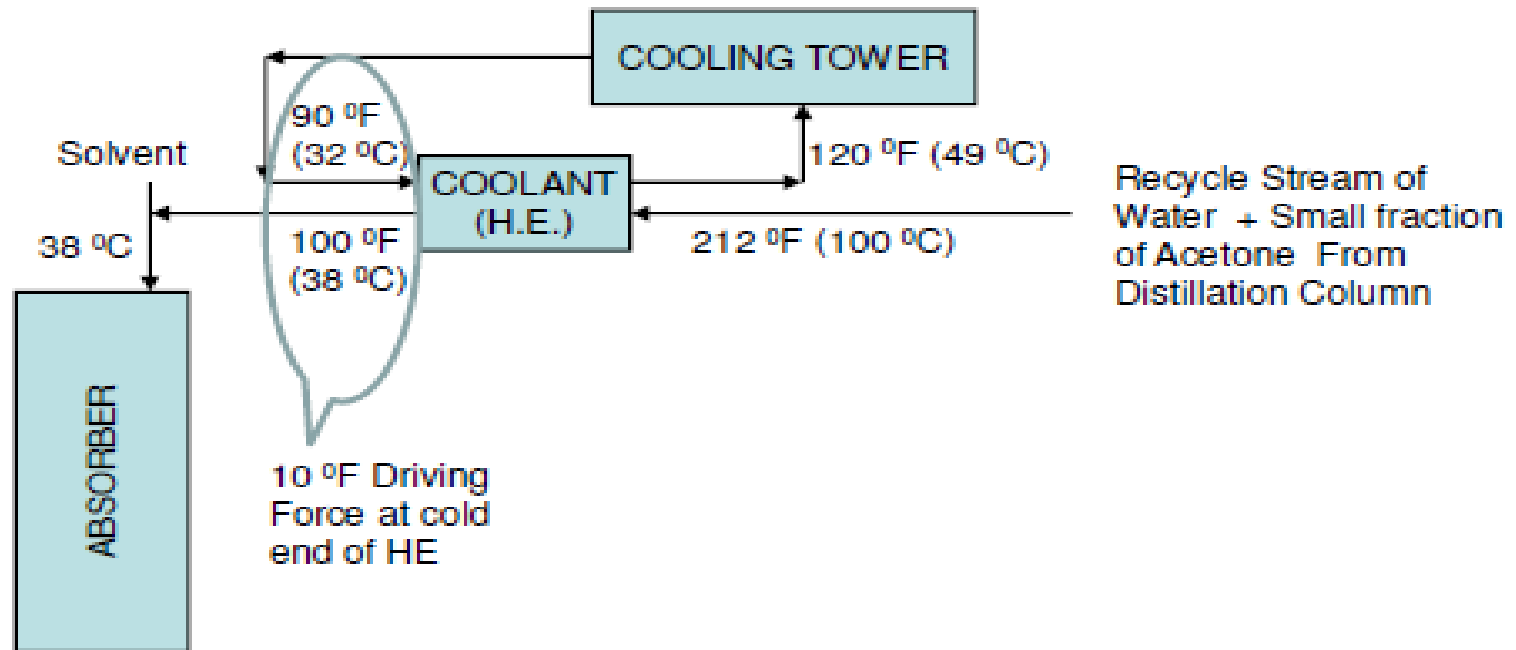


Fig. 3 Heat Exchange alternative in flow sheet



Design of Gas Absorber: Heuristic

- This reasoning is the basis for a design heuristic

H1: If a raw material component is used as the solvent (like water) in a gas absorber, consider feeding the process through the gas absorber



Design of Gas Absorber

- Considering the flow sheet shown in Fig. 1 because it is the simplest for further processing



Contd..

- In addition, we must evaluate whether we really want to use water as the solvent
- We arbitrarily choose to consider the flow sheet shown in Fig. 1 because it is the simplest for further processing



Design of Gas Absorber: Material Balances

- Identify the components that will appear in every stream
- The inlet gas flow to the absorber
 - 10.3 mol/hr of acetone + 687 mol/hr of air
- If we use well water as solvent
 - inlet solvent stream is pure water (100%, solute concentration is zero)

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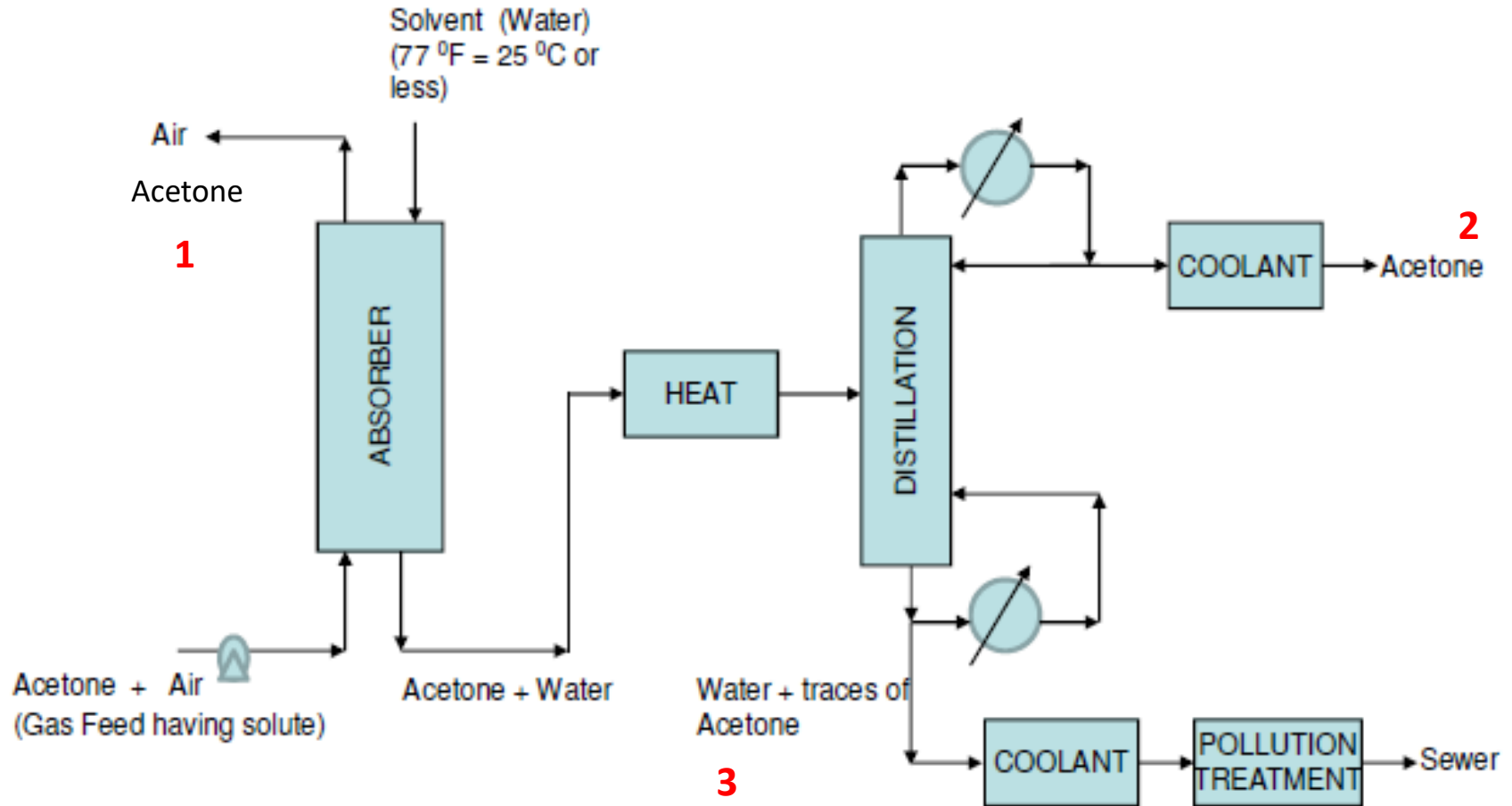


Fig. 1 Flow sheet of gas absorption

Douglas, J. M. Conceptual Design of Chemical Processes, 1988



Contd..

- The gas leaving the absorber (top) will contains
 - air, some acetone and some water
 - Since water is relatively inexpensive, neglecting this solvent loss

Specify Acetone Amount for Material Balance



- Specify the product specification in distillate overhead (2)
- Specify amount of acetone leaving in the other two streams (1 and 3)
- Recovery of 90, or 99, or 99.9% of the acetone in the gas absorber is possible?



Contd..

- Of course we can recover 90, or 99, or 99.9% or whatever of the acetone in the gas absorber
- Adding more trays to the top of absorber
 - The cost of the gas absorber will continue to increase as
- Increase the fractional recovery
 - but the value of the acetone lost to the flare system will continue to decrease
- There is a trade-off between these two, and
- Thus there is an optimal fractional recovery

Contd..

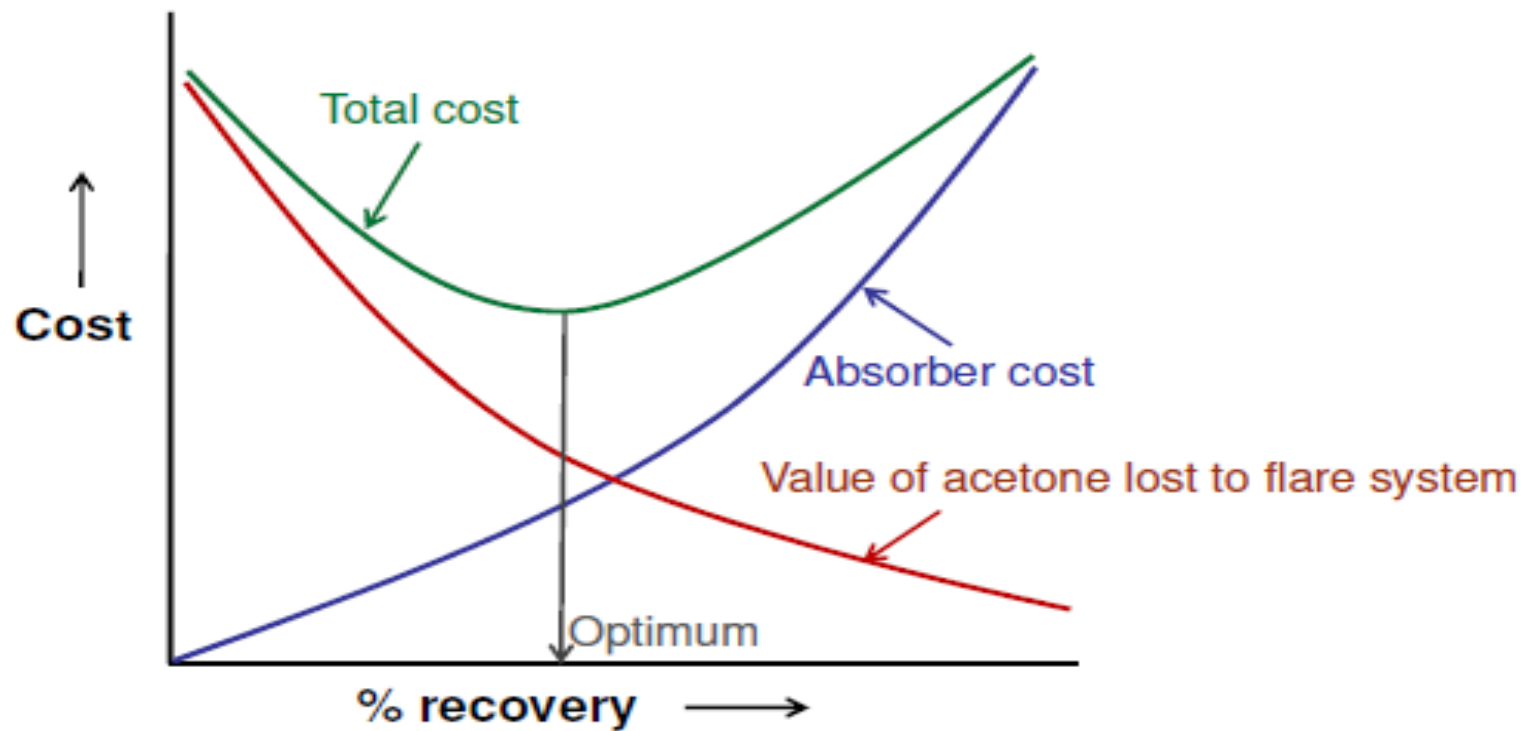


Fig.: %Recovery vs. Cost in Gas Absorber



Contd..

- There is optimum fractional recovery of bottoms in the distillation column
- As we add more & more plates in stripping section (bottom of distillation) of this column,
 - the still cost increases, but the value of the acetone lost to the sewer decreases

Contd..

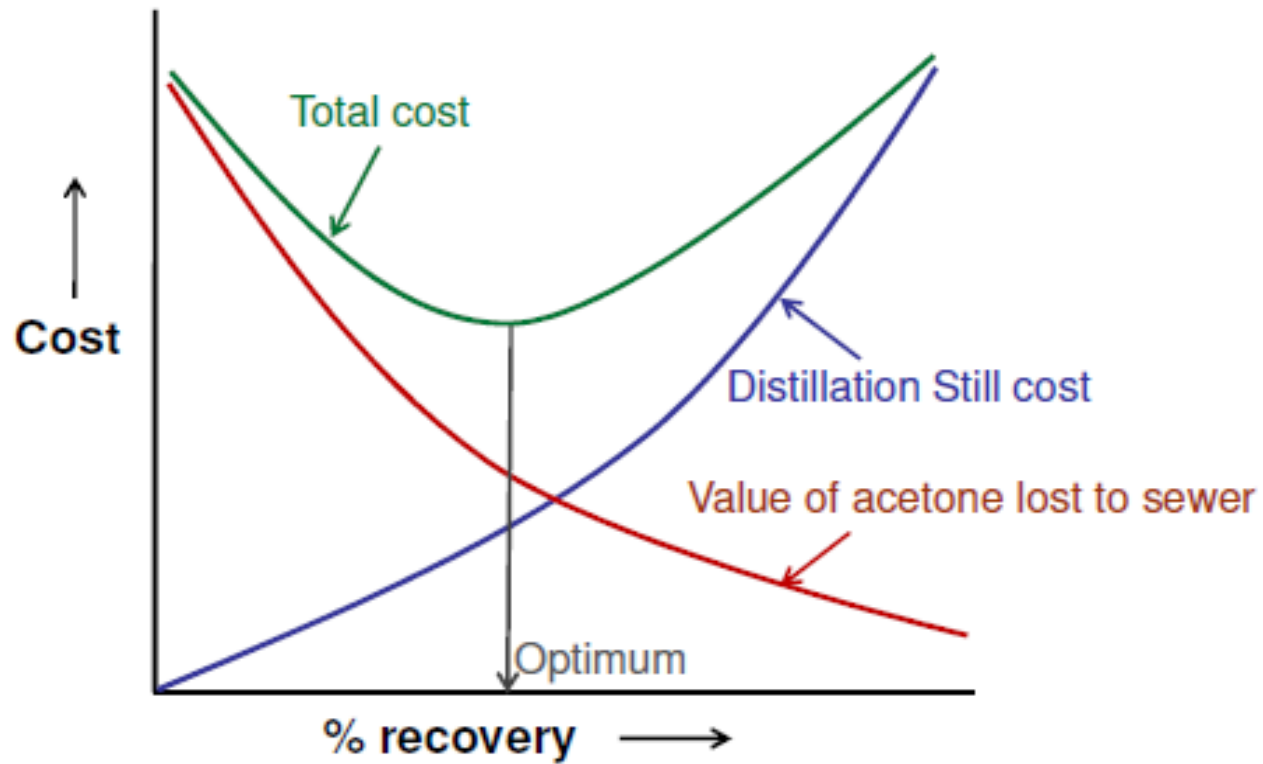


Fig.: %Recovery vs. Cost in Distillation

Design of Gas Absorber: Heuristics



H2: It is desirable to recover more than 99%
of all valuable materials
(we normally use 99.5% recovery as a first guess)



Design of Gas Absorber: Heuristics

H3: For an isothermal, dilute absorber, choose the solvent flow rate (L), such that $L = 1.4mG$
where, m = slope of equilibrium line, and G = gas molar flow rate (mol/hr)



Material Balances

- For the acetone-water system at 77°F (25°C) and 1 atm
 - Activity coefficient, $\gamma_s = 6.7$
 - Vapour pressure of acetone in air, $P_s^0 = 229$ mm Hg.
 - Air flow rate, $G = 687$ mol/hr

$$\begin{aligned}
 \phi y_s P_T &= \gamma x_s P_v \\
 m = \frac{y_s}{x_s} &= \frac{\gamma P_s^0}{P_T} = \frac{6.7(229)}{760} = 2.02
 \end{aligned}$$

Fugacity coefficient ϕ , Mol fraction of solute in gas y_s , Total pressure P_T , Activity coefficient γ , Mol fraction of solute in solvent x_s , Vapor pressure P_v , Vapour pressure of acetone in air P_s^0 , Air flow rate G .



Contd..

- Solvent flow rate (L) = $1.4 \text{ mG} = 1.4 \times 2.02 \times 687$
 $= \underline{1943 \text{ mol/hr}}$
- For a 99.5% recovery of acetone in the gas absorber,
 - The acetone lost from top of absorber = $0.005 (10.3) = \underline{0.05 \text{ mol/hr}}$
- And the acetone flow to the distillation column,
 $0.995 (10.3) = \underline{10.25 \text{ mol/hr}}$
- If 99.5% of acetone entering the still is recovered overhead,
 - Then acetone as distillate = $0.995 (10.25) = \underline{10.20 \text{ mol/hr}}$



Contd..

- Also if the product composition of acetone is specified to be 99%,
 - Then the amount of water in the product stream (distillate) will be

$$\left(\frac{1 - 0.99}{0.99} \right) (10.20) = 0.10 \text{ mol/hr}$$

- Then the bottom flows of acetone and water are
 - Acetone: $0.005 (10.25) = \underline{0.05 \text{ mol/hr}}$
 - Water: $1943 - 0.1 = \underline{1942.9 \text{ mol/hr}}$

Material Balances: Flowsheet

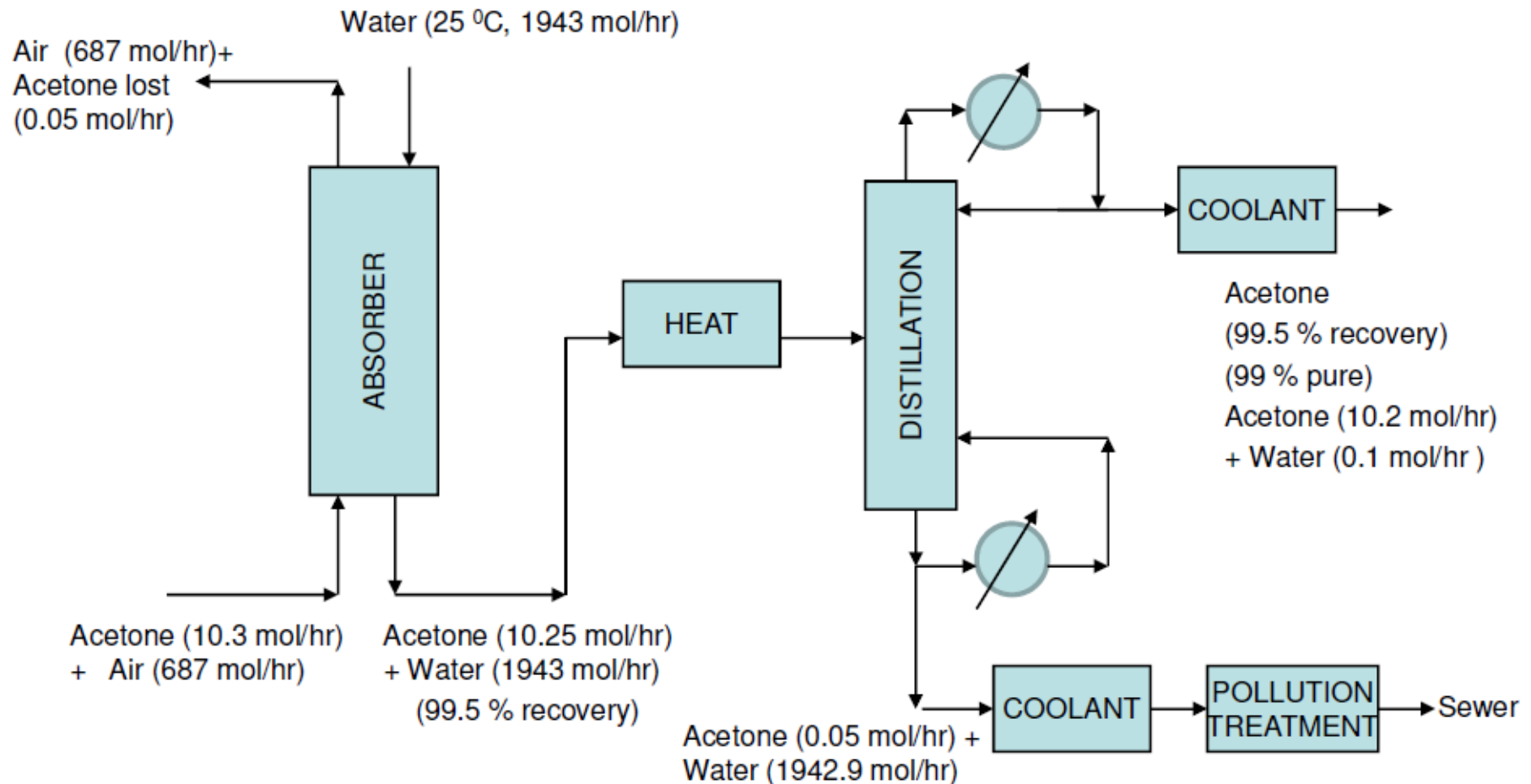


Fig.: Stream compositions and flow rates



Stream Cost Calculation: Acetone-Water System

- For acetone water system, with no recycling and 99.5 % recoveries

Acetone loss in absorber overhead (assume \$0.27/lb of Acetone)

$$\begin{aligned} &= (\$0.27/\text{lb}) \times (58 \text{ lb/mol}) \times (0.0515 \text{ mol/hr}) \times (8150 \text{ hr/yr}) \\ &= \$6600/\text{yr} \end{aligned}$$

Acetone loss in still bottom

$$\begin{aligned} &= (\$0.27/\text{lb}) \times (58 \text{ lb/mol}) \times (0.05 \text{ mol/hr}) \times (8150 \text{ hr/yr}) \\ &= \$6600/\text{yr} \end{aligned}$$



Contd..

Pollution treatment cost (assume \$0.25/lb BOD and 1 lb acetone/lb BOD)

$$\begin{aligned} &= (\$0.25/\text{lb BOD}) \times (1 \text{ lb BOD}/1 \text{ lb acetone}) \times (58 \text{ lb/mol}) \\ &\quad \times (0.05 \text{ mol/hr}) \times (8150 \text{ hr/yr}) \\ &= \$6100/\text{yr} \end{aligned}$$

Sewer charges (assume \$0.20/1000 gal)

$$\begin{aligned} &= (\$0.20/1000 \text{ gal}) \times (1 \text{ gal}/8.34 \text{ lb}) \times (18 \text{ lb/mol}) \times \\ &\quad (1942.9 \text{ mol/hr}) \times (8150 \text{ hr/yr}) \\ &= \$6800/\text{yr} \end{aligned}$$



Contd..

Solvent water (assume \$0.75/1000 gal)

$= (\$0.75/1000 \text{ gal}) (1 \text{ gal}/8.34 \text{ lb}) (18 \text{ lb/mol}) (1943 \text{ mol/hr}) (8150 \text{ hr/yr})$

$= \$25,600/\text{yr}$

- Each of these costs all together is essentially negligible compared to economic potential of **$\$1.315 \times 10^6/\text{yr}$** ,
 - We want to continue developing the design



Solvent Loss Calculations: Other than Water as Solvent

- For a low pressure absorber, fugacity correction factors are negligible
- Vapor-liquid equilibrium relationship for the solvent can be written as

$$P_T y_s = \gamma_s P_s^0 x_s$$

- With greater than 99% recovery of the solute, $x_s \approx 1$

Alternate Flow sheet: Using Solvent Recycle (Other than Water as solvent)

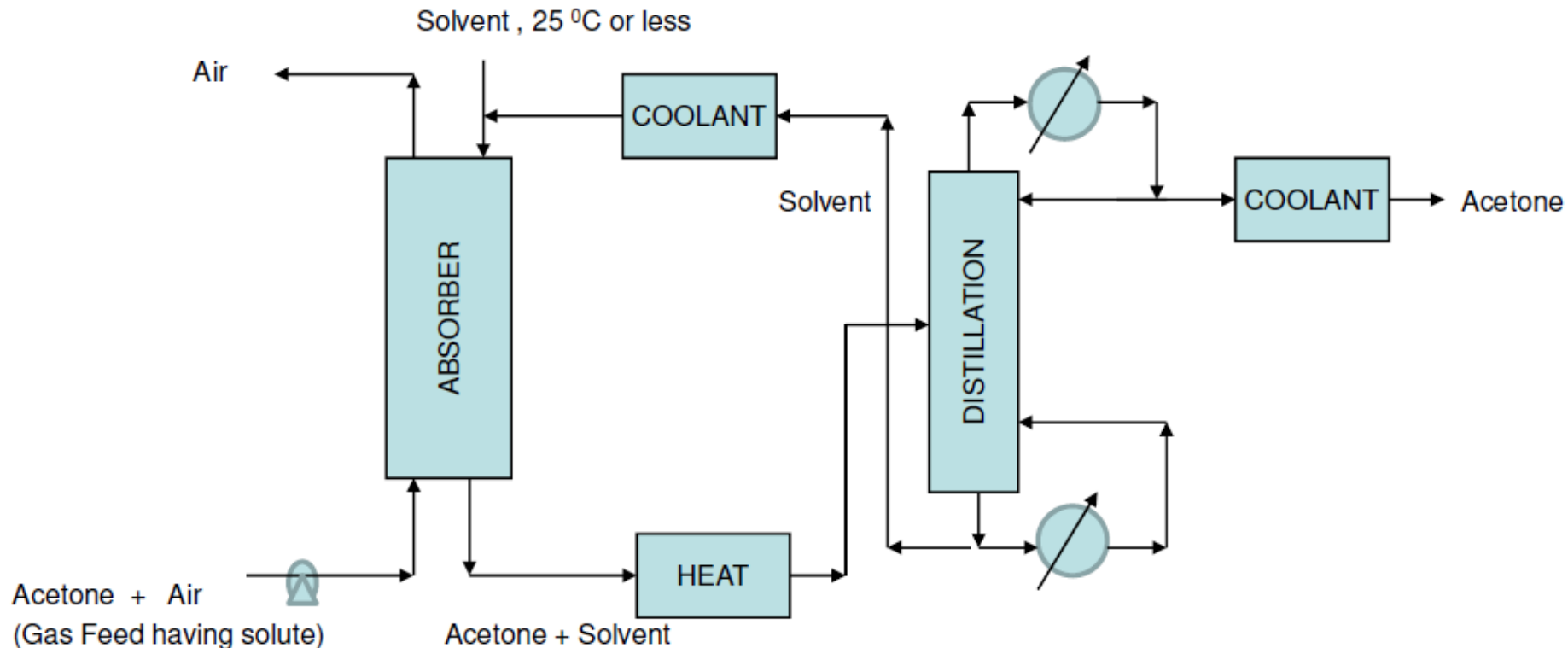


Fig.: Solvent Recycle to Gas-Absorber



Contd..

- If a solvent is used that is in the **homologous series** with the solute, then $\gamma_s = 1$

- Thus, from

$$P_T y_s = \gamma_s P_s^0 x_s$$

$$y_s = \frac{P_s^0}{P_T}$$



Contd..

Homologous series

- A series of chemical compounds of
 - (1) Uniform chemical type
 - (2) Showing a regular graduation in physical properties and
 - (3) Capable of being represented by a general molecular formula
 - e.g. alkanes: C_nH_{2n+2} (CH_4 , C_2H_6 , C_3H_8 , etc.)
 - Ketone: $C_nH_{2n}O$ (acetone, C_3H_6O , MIBK, $C_6H_{12}O$)



Cost of Solvent Loss Using MIBK as Solvent

- Suppose we consider using MIBK (Methyl Isobutyl Ketone) as a solvent and we recycle the MIBK
 - At 25°C, $P_T = 1 \text{ atm}$, $P_s^0 = 0.0237 \text{ atm}$
 - $y_s = P_s^0 / P_T = 0.0237 / 1 = 0.0237$

$$y_s = \frac{P_s^0}{P_T} = \frac{0.0237}{1} = 0.0237$$

- Therefore, solvent lost

$$m_s = y_s G = 0.0237 \times 687 \text{ mol/hr} = 19.7169 \text{ mol/hr}$$



Contd..

- MIBK lost (assume \$0.35/lb of MIBK = \$35/mol of MIBK)
 - (\$35/mol) (19.7169 mol/hr) (8150 hr/yr)
= \$4.464×10⁶/yr
- This value is much higher than E.P. (= \$1.315×10⁶)
 - So we drop any idea of using MIBK as solvent



Outline

- Design of Gas Absorber
 - Energy Balances

Material Balances

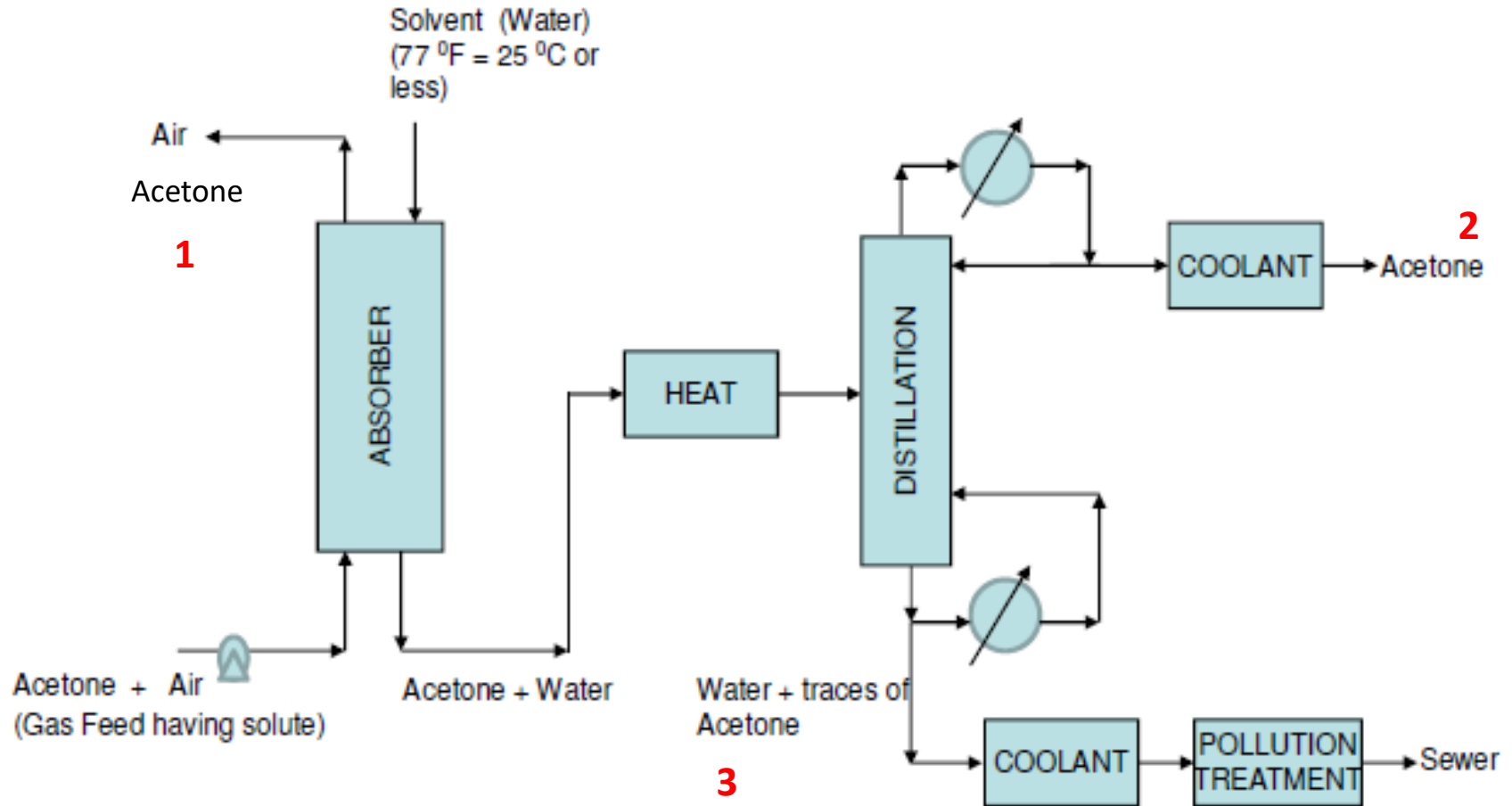


Fig. 1 Flow sheet of gas absorption

Douglas, J. M. Conceptual Design of Chemical Processes, 1988



Energy Balances for the Acetone Absorber

- Since the inlet composition to the gas absorber is quite dilute (10.3/687) (i.e. Acetone/Air)
 - assume that the absorber will operate isothermally (constant temperature)

Contd..

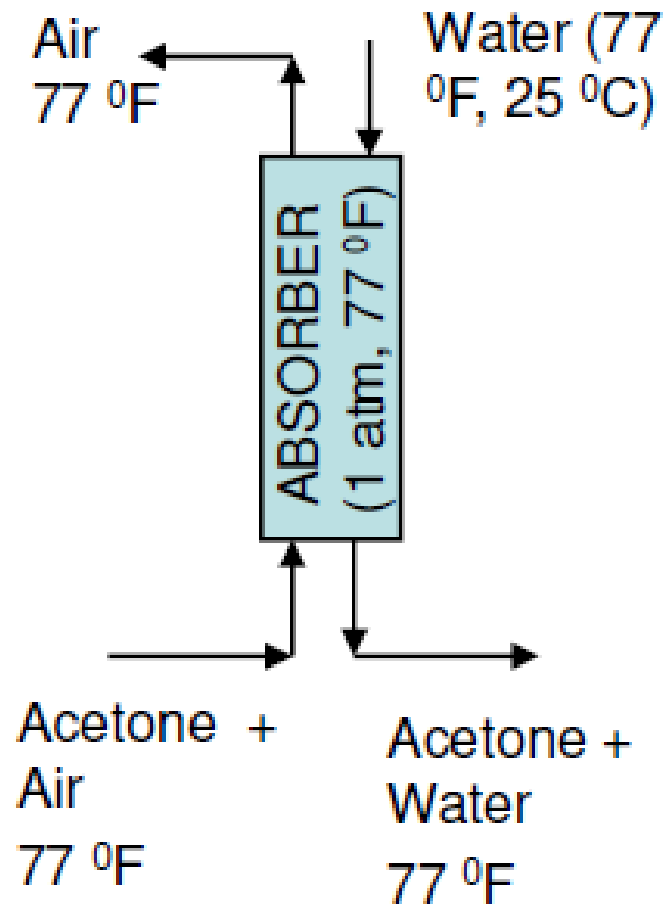


Fig.: Stream Temperatures in Gas absorber



Contd..

- Do not store our product stream (top product from Distillation column, Acetone) at its boiling point
 - so install a product cooler.
 - the temperature of the product stream leaving the product cooler will be 100 °F.
- Acetone product (99 % pure) contains 1 % water.
 - guess that the temperature of the overhead is essentially the same as the boiling point of acetone (56.5 °C or 135 °F)

Contd..

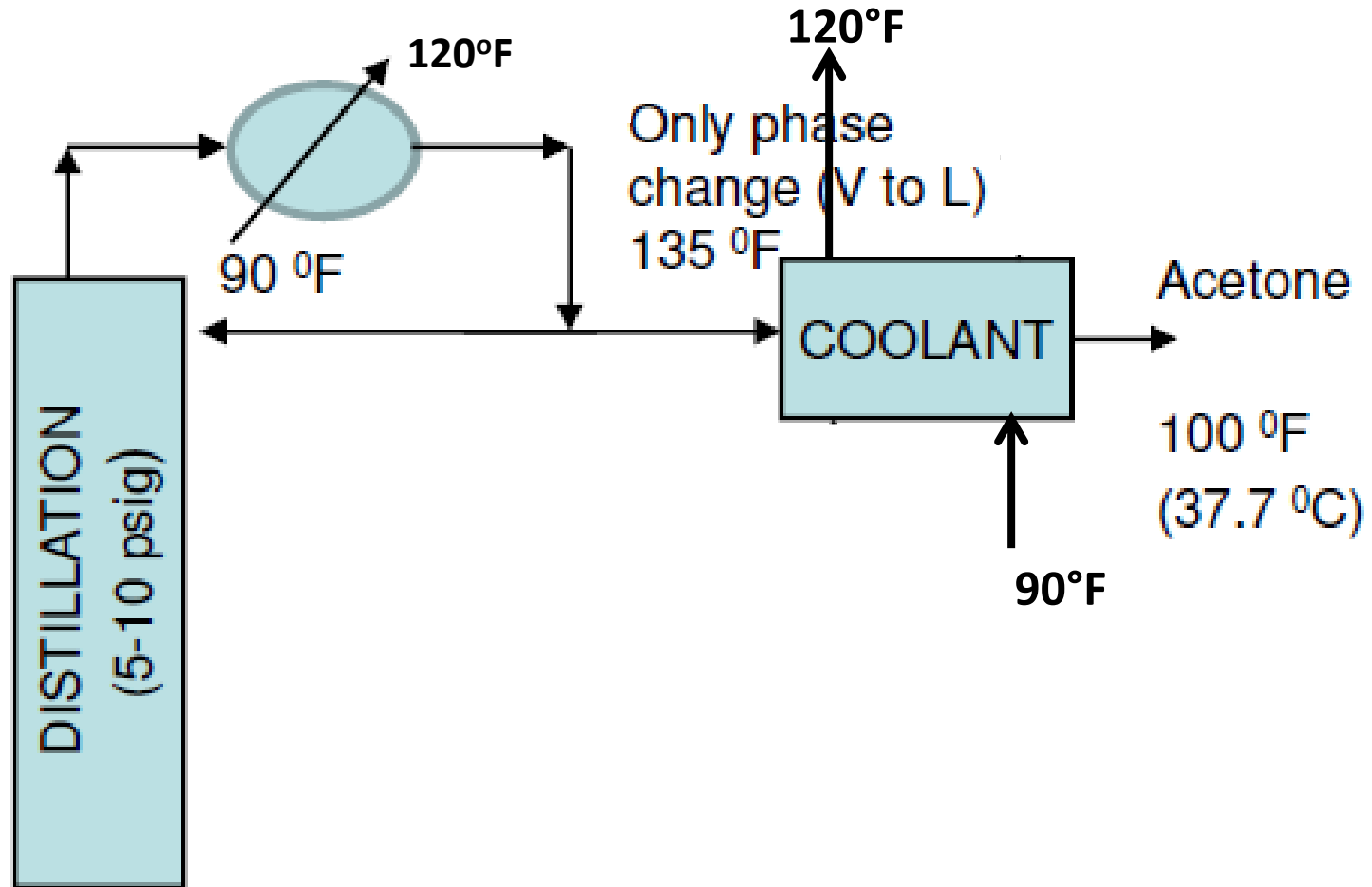


Fig.: Stream Temperatures in Still overhead



Contd..

- Similarly, assume that the bottom stream from the still is 212°F (i.e. B. P. of water = 100°C)
- Cool this waste stream to 100°F (cooling water temperature) prior to pollution treatment.

Contd..

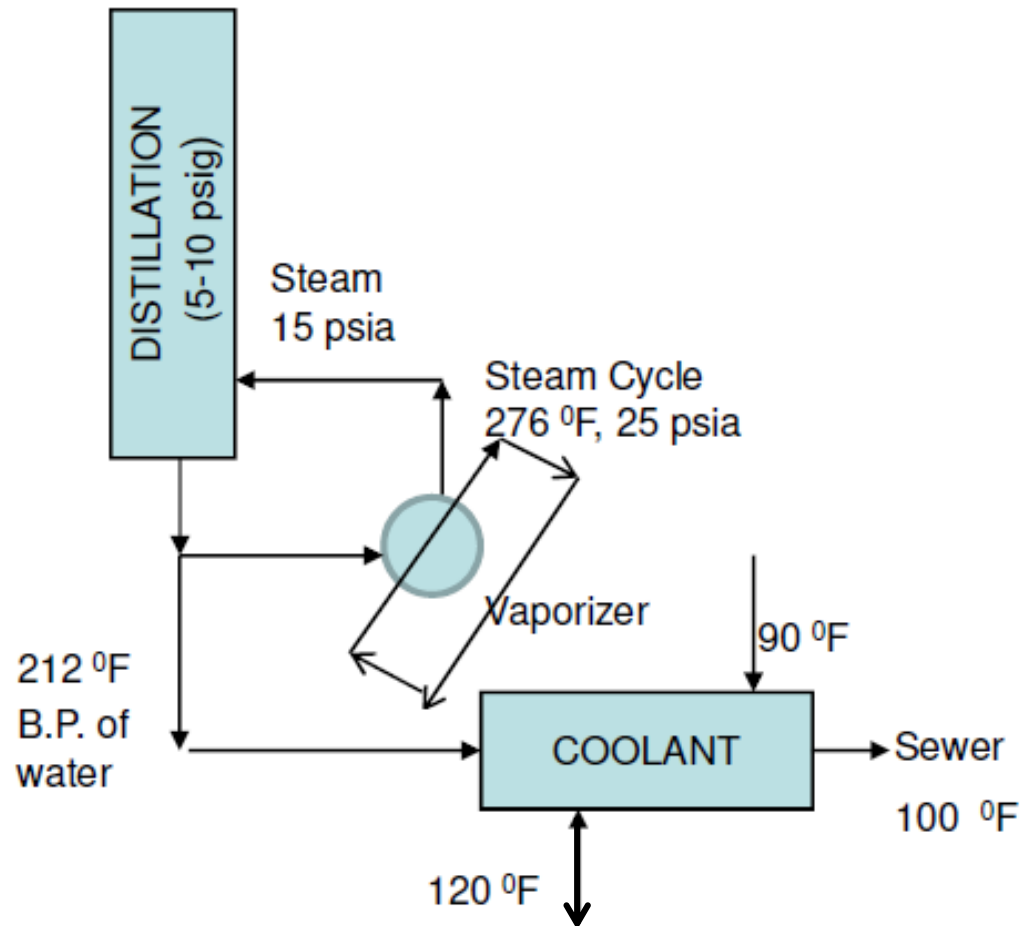


Fig.: Stream Temperatures at Still bottom

Energy Integration Flowsheet

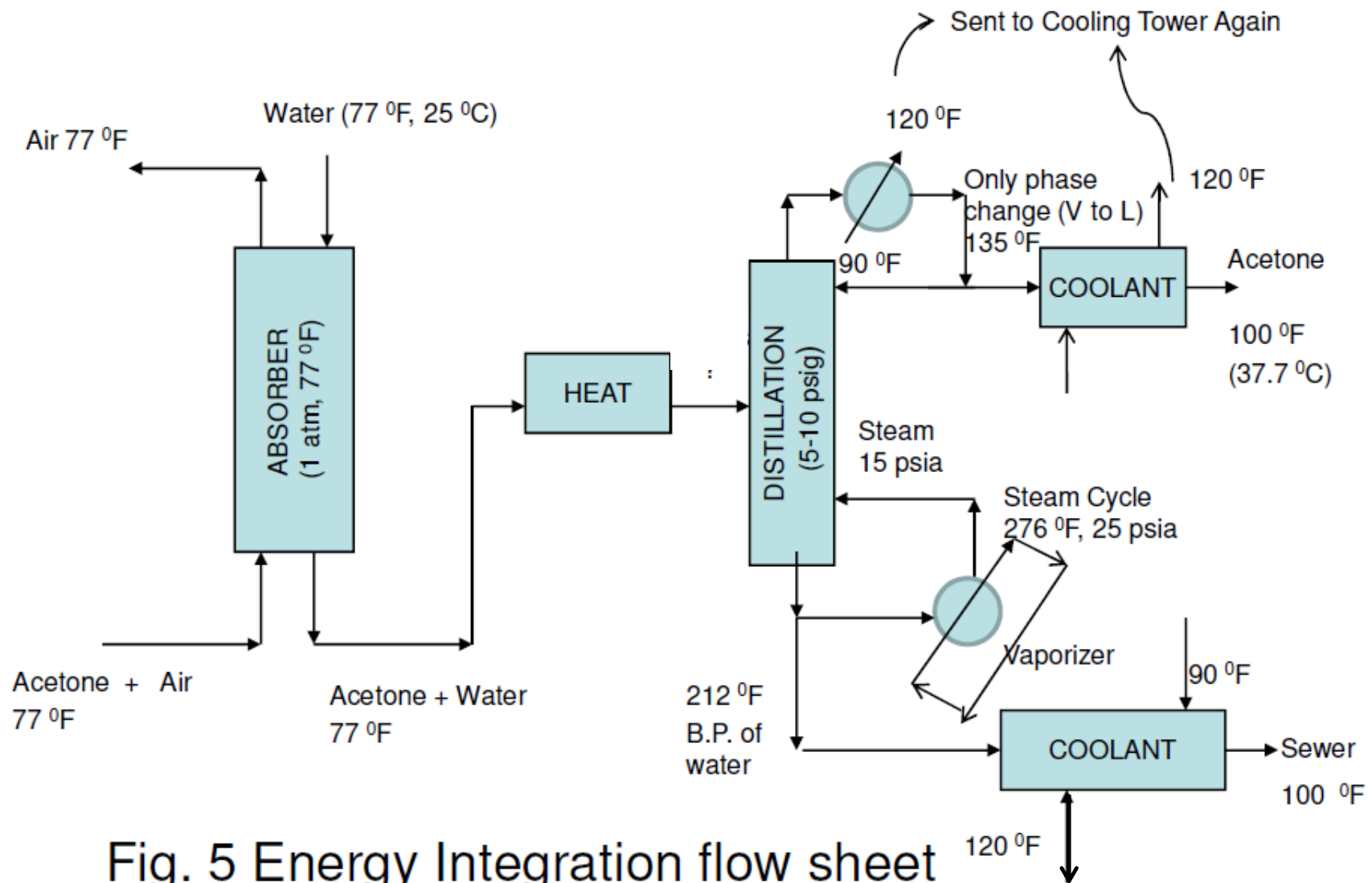


Fig. 5 Energy Integration flow sheet

Energy Balances for the Acetone Absorber



- Must specify the temperature of the stream entering the distillation column



Contd..

- If we do not preheat the feed stream entering the distillation column to close the saturated liquid condition,

What will be consequences?



Contd..

- Energy Balances

- With the specified stream temperatures and estimated stream flows, heat loads of various streams can be calculated

$$Q_i = F_i C_{Pi} (T_{in} - T_{out})$$

- Thus we can decide on HEN and calculate
 - The H.E. areas
 - Annualized H.E. capital costs and
 - The utility costs



Outline

- Equipment Design Consideration
 - Number of plates in gas absorber
 - Cause-and-effect relationship of process design variables
 - Simplifying unit-operation models (Back-of-the-Envelope design equation)
- Rules of thumb: Liquid flow rate to absorber



Equipment Design Consideration

- Calculate the size & cost of the absorber and distillation column
- Need to understand the cause-and-effect relationships (Input-output models) of the design variables
 - System vs. Unit Approach



Gas Absorber

- For isothermal dilute system, the **Kremser's Eqn.**

$$N + 1 = \frac{\ln \left[\left(\frac{L}{mG} - 1 \right) \left(\frac{y_{in} - mx_{in}}{y_{out} - mx_{in}} \right) + 1 \right]}{\ln \left(\frac{L}{mG} \right)}$$

- Pure water** as the solvent,

$$x_{in} = 0$$



Contd..

- From the rules of thumb, discussed earlier,

$$y_{out} \leq (1 - 0.99)y_{in}$$

$$\left\{ \begin{array}{l} \frac{y_{out}}{y_{in}} \leq (1 - 0.99) = 0.01 \\ L = 1.4mG = 1.4 \left(\frac{\gamma P^o}{P_T} \right) G \end{array} \right\}$$



Effect of Design Variables: Column Pressure

- If we double the column pressure (P_T),
- L decreases by a factor of 2,
- but since $L/mG = 1.4$, i.e. constant
- both L and m are $= f(P_T)$, decreases
- The number of plates required in gas absorber does not change.

$$L = 1.4mG = 1.4 \left(\frac{\gamma P^\circ}{P_T} \right) G$$

$$N + 1 = \frac{\ln \left[\left(\frac{L}{mG} - 1 \right) \left(\frac{y_{in} - mx_{in}}{y_{out} - mx_{in}} \right) + 1 \right]}{\ln \left(\frac{L}{mG} \right)}$$



Contd..

Lower values of L means

1. D.C. feed will be more concentrated
2. The reflux ratio decreases
3. The vapor rate in the still decreases
4. The column diameter decreases
5. Sizes of condenser and reboiler decreases (load decreases)
6. Steam and cooling water requirement decreases



Is there any consequence of increasing the pressure?



Effect Design Variables: Solvent MIBK

- For MIBK
- $\gamma = 1$ in place of 6.7 in case of water
- Liquid rate could be decreased as m will decrease
- Decreases the D.C. cost
- No. of plates in absorber will not change as L/mG is constant



Effect of Design Variables: Operating Temperatures

- If we change the inlet water (solvent to absorber) temperature to 40°C (112 °F),

$$\gamma = 7.8 \text{ \& } P^0 = 421 \text{ mm Hg, } P^0 \propto T, \gamma \propto T$$

- Thus ' m ' increases , L increases (So the D. C. Cost increases)
- But number of trays in absorber does not change ($L/mG = \text{Const}$)



Simplifying unit-operation models

- Also called as Back-of-the-Envelope Design equation
- Significance and order of magnitude of various terms in Kremser's Eqn.

$$N + 1 = \frac{\ln \left[\left(\frac{L}{mG} - 1 \right) \left(\frac{(y_{in} - mx_{in})}{y_{out} - mx_{in}} \right) + 1 \right]}{\ln \left(\frac{L}{mG} \right)}$$



Contd..

$$N + 1 = \frac{\ln \left[\left(\frac{L}{mG} - 1 \right) \left(\frac{(y_{in} - mx_{in})}{y_{out} - mx_{in}} \right) + 1 \right]}{\ln \left(\frac{L}{mG} \right)}$$

L.H.S. of Kremser Eqn.

$$= N + 1$$

Assuming N @ =15-20 trays and 10% error is allowed

$$N + 1 \approx N$$

➤ L.H.S of Kremser Eqn. = N



Contd..

$$N + 1 = \frac{\ln \left[\left(\frac{L}{mG} - 1 \right) \left(\frac{y_{in} - mx_{in}}{y_{out} - mx_{in}} \right) + 1 \right]}{\ln \left(\frac{L}{mG} \right)}$$

R.H.S.

For pure solvents, $x_{in} = 0$

(Solute concentration in pure solvent = 0)

Numerator of R.H.S. of Kremser Eqn.

$$= \ln \left\{ \left(\frac{L}{mG} - 1 \right) \frac{y_{in}}{y_{out}} + 1 \right\}$$



Contd..

Rules of thumb indicate

$$\frac{L}{mG} \cong 1.4 \ \& \ \frac{y_{in}}{y_{out}} \cong 100$$

Thus

$$\left\{ \left(\frac{L}{mG} - 1 \right) \frac{y_{in}}{y_{out}} + 1 \right\} \cong 40 + 1$$

$$1 \ll 40$$



Contd..

- Applying the order of magnitude criteria
($1 \ll 40$)

$$\ln \left\{ \left(\frac{L}{mG} - 1 \right) \left(\frac{y_{in}}{y_{out}} \right) + 1 \right\} \cong \ln \left\{ \left(\frac{L}{mG} - 1 \right) \left(\frac{y_{in}}{y_{out}} \right) \right\}$$

- The denominator of R.H.S. Kremser Eqn.

$$\ln \left(\frac{L}{mG} \right) = \ln (1 + \epsilon)$$

- From Taylor series expansion,

$$\ln (1 + \epsilon) \cong \epsilon$$



Contd..

$$\epsilon = \frac{L}{mG} - 1 = 1.4 - 1 = 0.4 \Rightarrow \ln \frac{L}{mG} = 0.4$$

- With these, simplification and replacing 'ln' by 'log' we get

$$N = \frac{\ln \left[0.4 \left(\frac{y_{in}}{y_{out}} \right) \right]}{0.4} = \frac{2.3 \log \left[0.4 \left(\frac{y_{in}}{y_{out}} \right) \right]}{0.4}$$



Contd..

$$\Rightarrow N = 6 \log \left[0.4 \left(\frac{y_{in}}{y_{out}} \right) \right] = 6 \left[\log (0.4) + \log \left(\frac{y_{in}}{y_{out}} \right) \right] = -2 + \log \left(\frac{y_{in}}{y_{out}} \right)$$

$$\Rightarrow N + 2 = 6 \log \left(\frac{y_{in}}{y_{out}} \right)$$

- Approximation for a recovery of 99% gives 10 trays instead of actual value of 10.1
- For recovery of 99.9% gives 16 trays which is a very good estimation



Rules of Thumb: Liquid Flow Rate to gas Absorbers

- For Isothermal, dilute gas absorbers
 - Kremser Eqn. can be used for calculating No. of trays reqd. (N) for a specified recovery as a function of L/mG

$$N + 1 = \frac{\ln \left[\left(\frac{L}{mG} - 1 \right) \left(\frac{y_{in} - mx_{in}}{y_{out} - mx_{in}} \right) + 1 \right]}{\ln \left(\frac{L}{mG} \right)}$$

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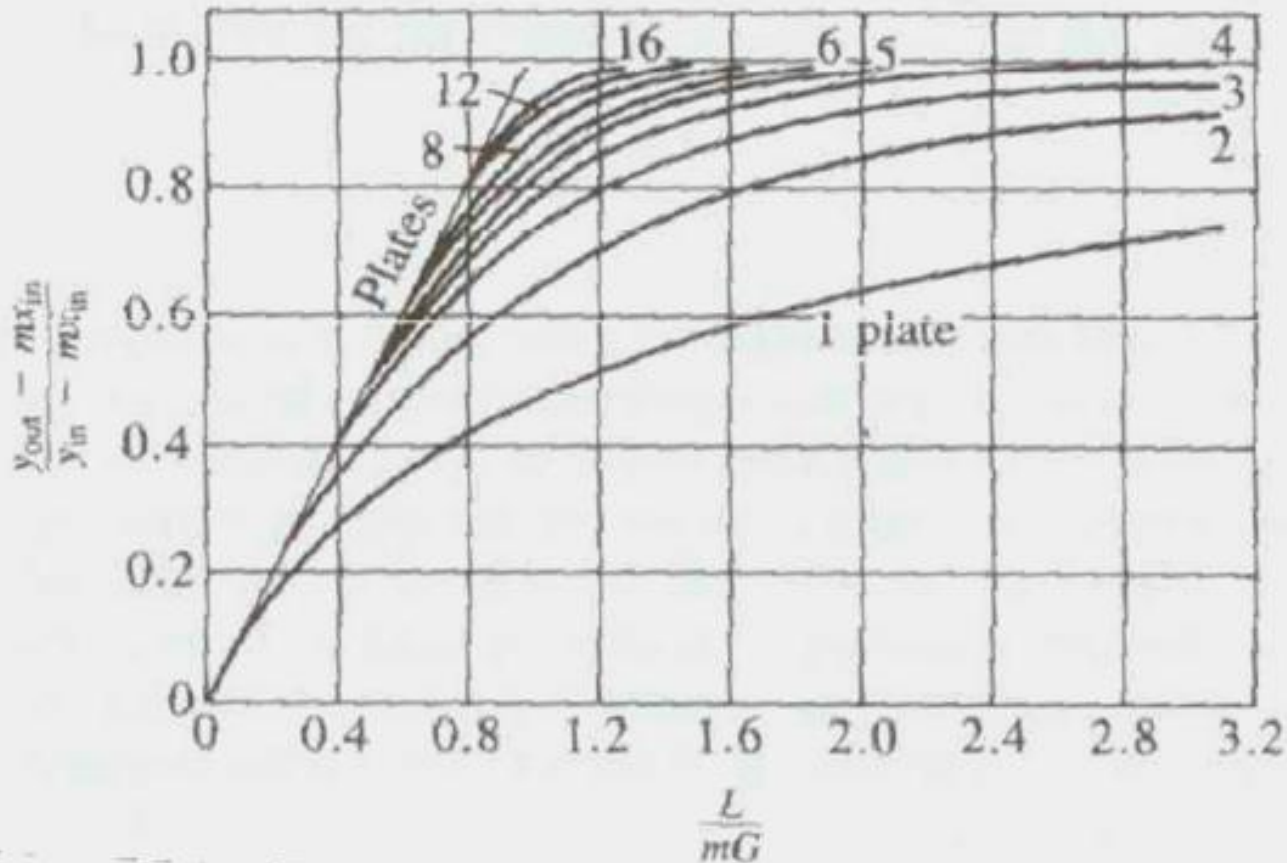


Fig.: Liquid Flow Rate vs. Fractional Recovery : Kremser Eqn.

Douglas, J. M. Conceptual Design of Chemical Processes, 1988, pp. 86



Contd..

$$\underline{L/mG < 1}$$

⇒ **INFINITE** No. of trays are required for near complete recovery (Infinite capital cost)

$$\underline{L/mG = 2}$$

⇒ **5 plates** are required for complete recovery ($\approx 100\%$)

– Large L correspond to dilute feeds to the distillation column



Contd..

$L/mG > 2$

⇒ We obtain tiny, inexpensive absorbers but very expensive D.C.

- **Based on above arguments, $1 < L/mG < 2$**

$L/mG = 1.5$: Observe the shape of the curves near L/mG

- Better trade off (with high recoveries)
 - Decreasing no. of trays in absorber (capital cost Vs. Increasing capital cost & operating cost of D.C.)
- Common Rule of Thumb: By decreasing L (such that $L/mG=1.4$) almost 100 % recovery



Rules of Thumb: Fractional recovery in Gas Absorbers (> 99%)

- For a fixed solvent flow rate,
 - increase the recovery of solute by simply adding trays in the gas absorber
- The trade-off is in between

Capital cost (the absorber cost)

vs.

Operating cost (the solute loss)



Cost Model (Fractional Recovery in Gas Absorber)

Total annual cost (TAC) model

$$\text{TAC} = (C_s \$ / \text{mol})(Gy_{\text{out}} \text{ mol/hr})(8150 \text{ hr/yr}) \\ + C_N [\$ / (\text{plate.yr})](N \text{ plates})$$



Cost model for Sensitivity Analysis

- Optimum Design : using simplified design (Kremser Eqn)

$$TAC = 8150 C_s G y_{in} \left(\frac{y_{out}}{y_{in}} \right) + C_N \left(6 \log \frac{y_{in}}{y_{out}} - 2 \right)$$

- For optimum fractional loss

$$\frac{\delta(TAC)}{\delta(y_{out} / y_{in})} = 0 = 8150 C_s G y_{in} - 6 \frac{C_N}{y_{out} / y_{in}}$$

$$\frac{y_{out}}{y_{in}} = 6 \frac{C_N}{8150 C_s G y_{in}}$$



Contd..

- Solute loss must be optimum
- Considering some typical values,
 $C_s = \$15.5/\text{mol}$
 $G_{in} = 10 \text{ mol/hr}$; $C_N = \$850/\text{plate yr}$
(Based on \$2550 per plate & a CCF of $1/3 \text{ yr}^{-1}$)

$$\frac{y_{out}}{y_{in}} = \frac{6(850)}{8150(15.5)(10)} = 0.004$$

- i.e. 0.4 % loss of solute
 - Which corresponds to, fractional recovery = 99.6 %



Sensitivity

- The significant feature of this elementary analysis is not the relationship for the optimum design

$$\frac{y_{out}}{y_{in}} = 6 \frac{C_N}{8150 C_s Gy_{in}}$$

- Change any of the number in the numerator or denominator by 100% (C_N , C_s or Gy_{in}) and the optimum fractional recovery is more than 99%



Contd..

- Thus, the result is very insensitive to any of the design or cost parameters
- There is no incentive for refining the cost data used in the analysis



Limitations of Rules of Thumb

- The rules of thumb
 1. $L/mG = 1.4$
 2. Fractional recovery $> 99 \%$
- Kremser Eqn. is valid only for
 - Isothermal, dilute systems (where both the operating & equilibrium lines are straight)



Contd..

- For a isothermal and dilute system
 - The minimum solvent flow rate corresponds to ∞ no. of trays
 - The condition at the bottom end of the tower when x_{out} of liquid is in the equilibrium with y_{in} of gas for a fixed fractional recovery
 - So that an ∞ no. of trays is required at the concentrated end of column

Contd..

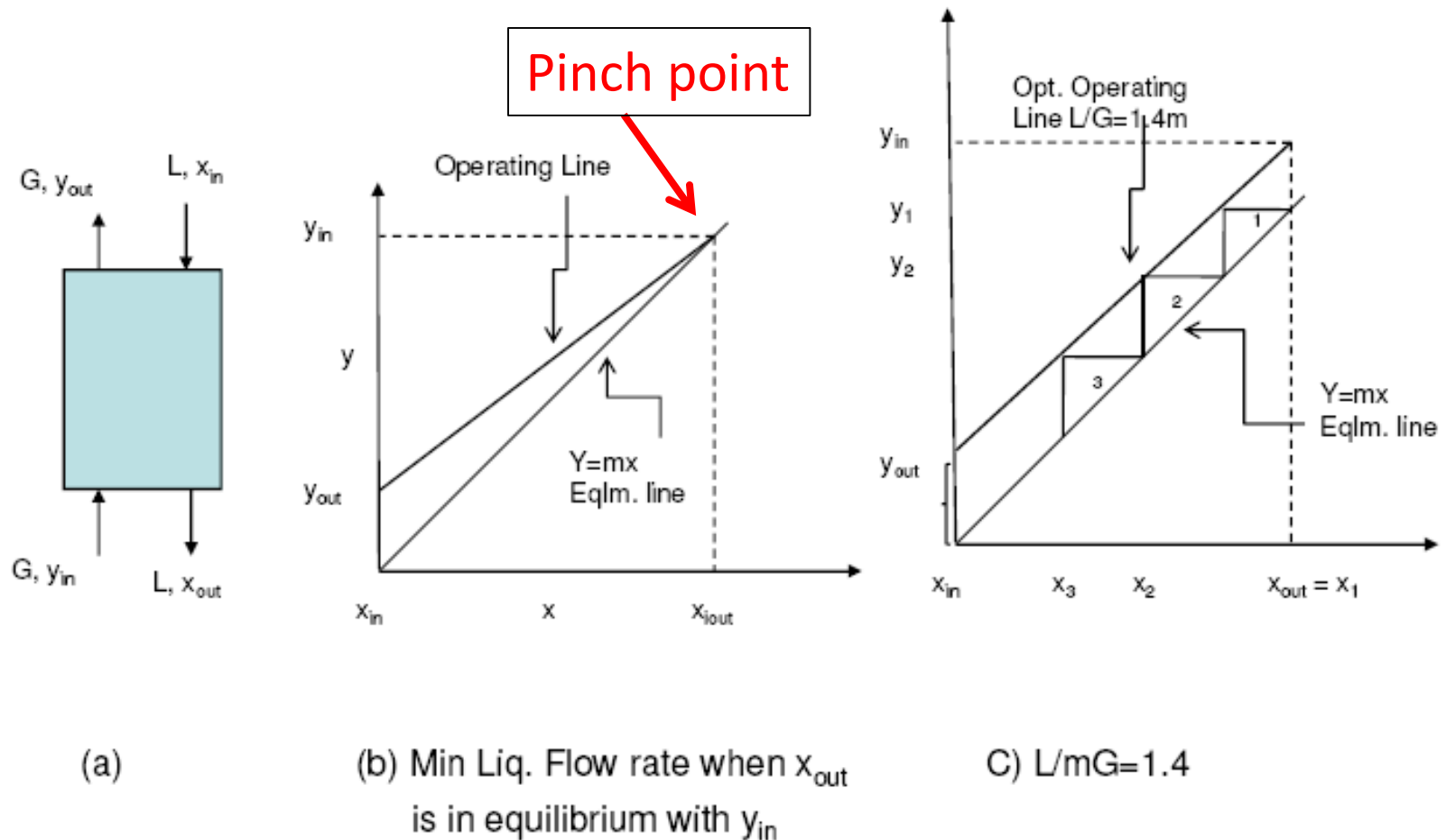


Fig.: Minimum Liquid Flow Rate - Isothermal

Douglas, J. M. Conceptual Design of Chemical Processes, 1988, pp. 89



Contd..

1. As per economic trade off:
Absorber to operate with $L=1.4 mG$ and a high fractional recovery ($> 99\%$)
2. From Fig., we see that these results correspond to almost a pinch zone at the top (dilute end) of the absorber

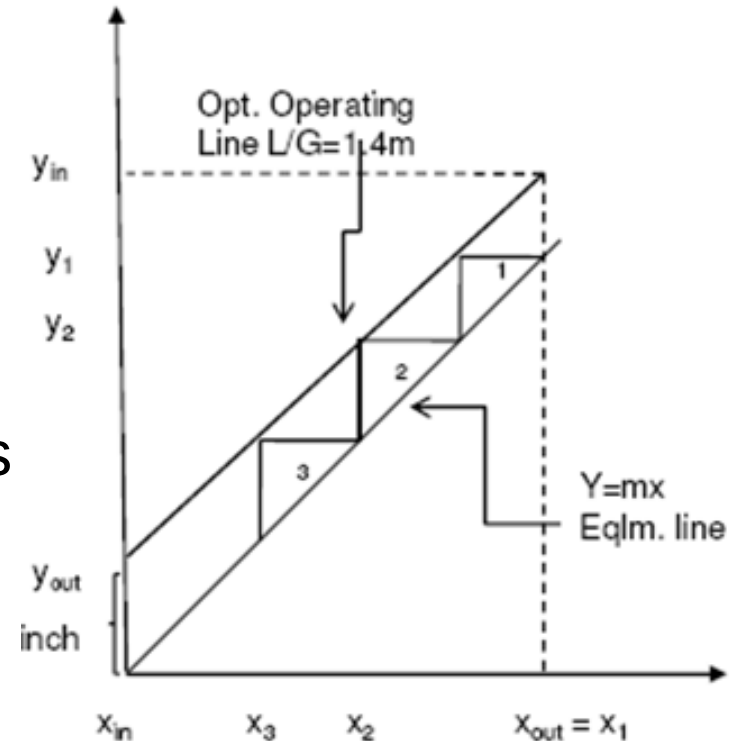


Fig.: Minimum Liquid Flow Rate - Isothermal

C) $L/mG=1.4$

Contd..

- For concentrated mixtures of solute

Douglas, J. M. Conceptual Design of Chemical Processes, 1988, pp. 89

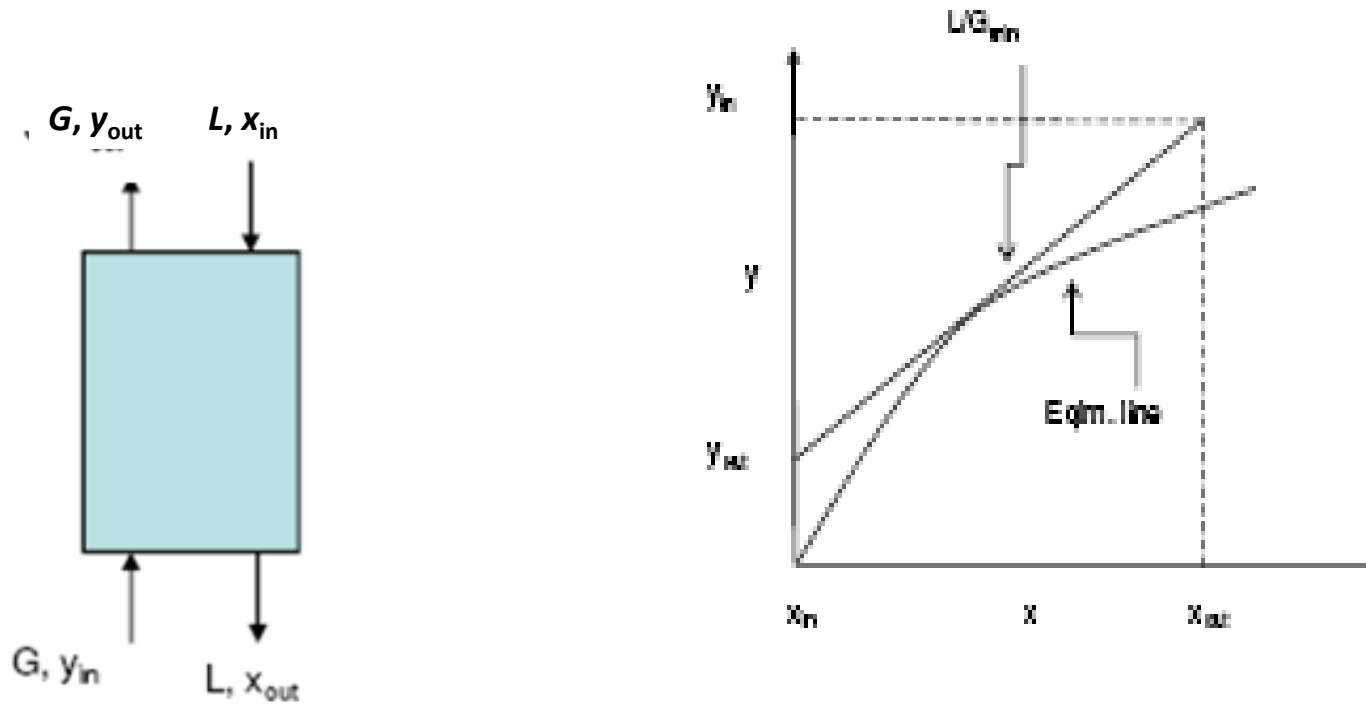


Fig. Minimum liquid flow rate- concentrated mixtures



Contd..

- As the liquid flow rate is increased
 - It will require more trays to get from y_{in} to y_{out} than a corresponding change for the dilute case
- For Concentrated Solute case, optimum condition

$$L/mG > 1.4, \% \text{ recovery} \leq 99\%$$



How do we design absorber column to work isothermally?



Contd..

For adiabatic absorbers

- As the solute leaves the gas stream and is taken up by the solvent, the solute gives up its heat of condensation
 - This heat effect causes the temperature of the liquid stream to increase as it approaches the bottom end of the tower
 - Increasing liquid temperatures increase the vapour pressure of the solute

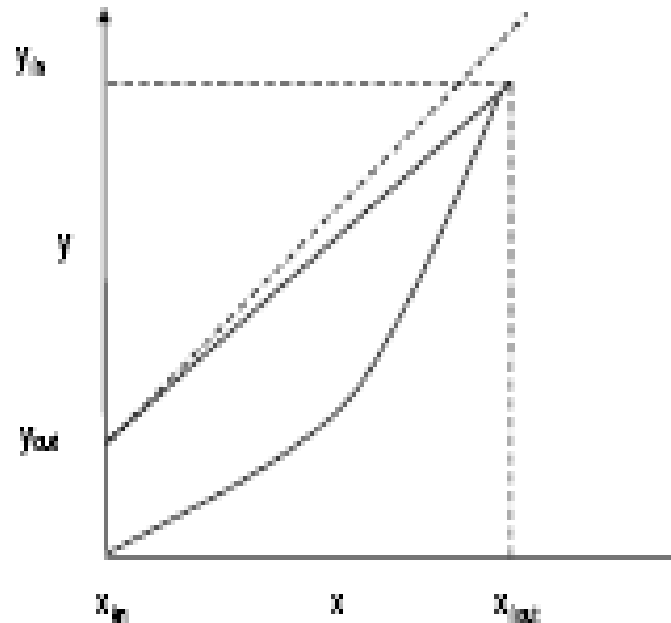


Contd..

- And as , $m = \frac{\gamma P_s^0}{P_T}$ the slope of equilibrium line (m) increases
- Thus, the minimum liquid flow rate occurs, when there is a pinch at the bottom of the tower when $y_{in} = m_{out} x_{out}$



Contd..



Slope of Equilibrium Line increases as liquid goes down the bottom of the absorber

Douglas, J. M. Conceptual Design of Chemical Processes, 1988, pp. 90



Contd..

- A very small increase in the solvent flow rate above the minimum, will allow us to get from y_{in} to y_{out} with a very few plates
- However, the upward curving nature of the equilibrium line means
 - that the minimum solvent flow rate will be much larger



Contd..

- The minimum liquid flow rate for an adiabatic absorber may be 10 times more than the rule of thumb value i.e. $L=1.4 \text{ mG}$ based on the inlet liquid temperature
 - i.e. $L= 14 \text{ mG}$ for adiabatic absorber
- Indiscriminate use of rules of thumb may lead to an inoperable design.
- In general
 - Every thumb rule has some limitations



Contd..

- Obtain only a few plates (small absorber cost) and large liquid flows (large still costs) which is not a desirable situation
- Thus, it is common practice to put cooling coils, or one or more pump-around cooling loops on the bottom 2 or 3 trays of a gas absorber to force it to behave more as an isothermal tower



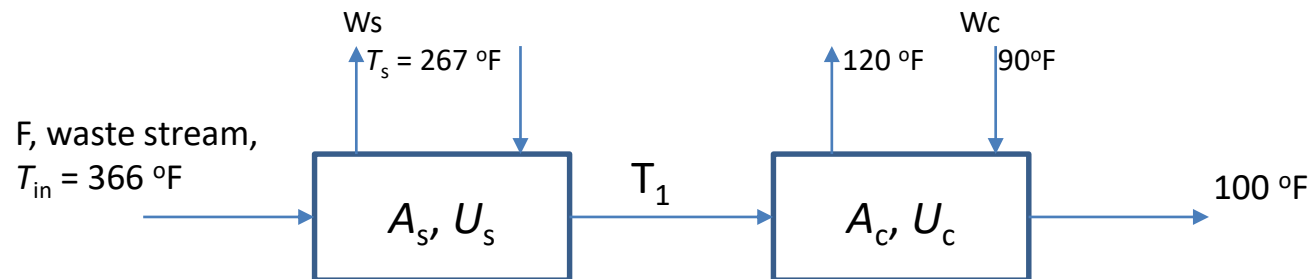
Heuristic

H4: AVOID THE USE OF ADIABATIC ABSORBERS
(Unless there is only a small temperature rise across the absorbers)



Problem

A rule of thumb commonly used in design is that the approach temperature in a heat exchanger should be 10 °F in the range from ambient to the boiling point of organics. To evaluate the heuristic, consider the simple system shown in Fig., where we are attempting to recover some heat from a waste stream by producing steam. Calculate these bounds for a case where,





F = Mass flow rate = 51000 lb/h;

C_p = heat capacity = 1.0 Btu/lb °F;

U_c = Overall heat transfer coefficient of coolant = 30 Btu/h.ft².°F;

U_s = Overall heat transfer coefficient of steam = 20 Btu/h.ft².°F;

C_A = Annual cost of heat exchanger per unit area = \$11.38 /yr.ft²;

C_w = Annual cost of cooling water = \$0.07388 / [lb/hr]yr;

C_s = Value of steam produced = \$21.22 /[lb/h]yr;

ΔH_s = latent heat of vaporization of steam = 933.7 Btu/lb;

T_s = Temperature of steam = 267 °F;

T_{in} = Inlet temperature = 366 °F



Problem

The minimum number of trays in a distillation column for a binary mixture is estimated by taking the sum of the boiling points and dividing by 3 times their difference. Show that the back-of-envelope model is essentially equivalent to Fenske's equation for the minimum number of trays.



Thank You